

How Modern Browsers Work

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Web developers often treat the browser as a black box that magically transforms HTML, CSS, and JavaScript into interactive web applications. This article dives into how modern browsers work — focusing on Chromium's architecture and internals, while noting where other engines differ. From the networking stack and parsing pipeline to the rendering process via Blink, JavaScript engine via V8, module loading, multi-process architecture, security sandboxing, and developer tooling.

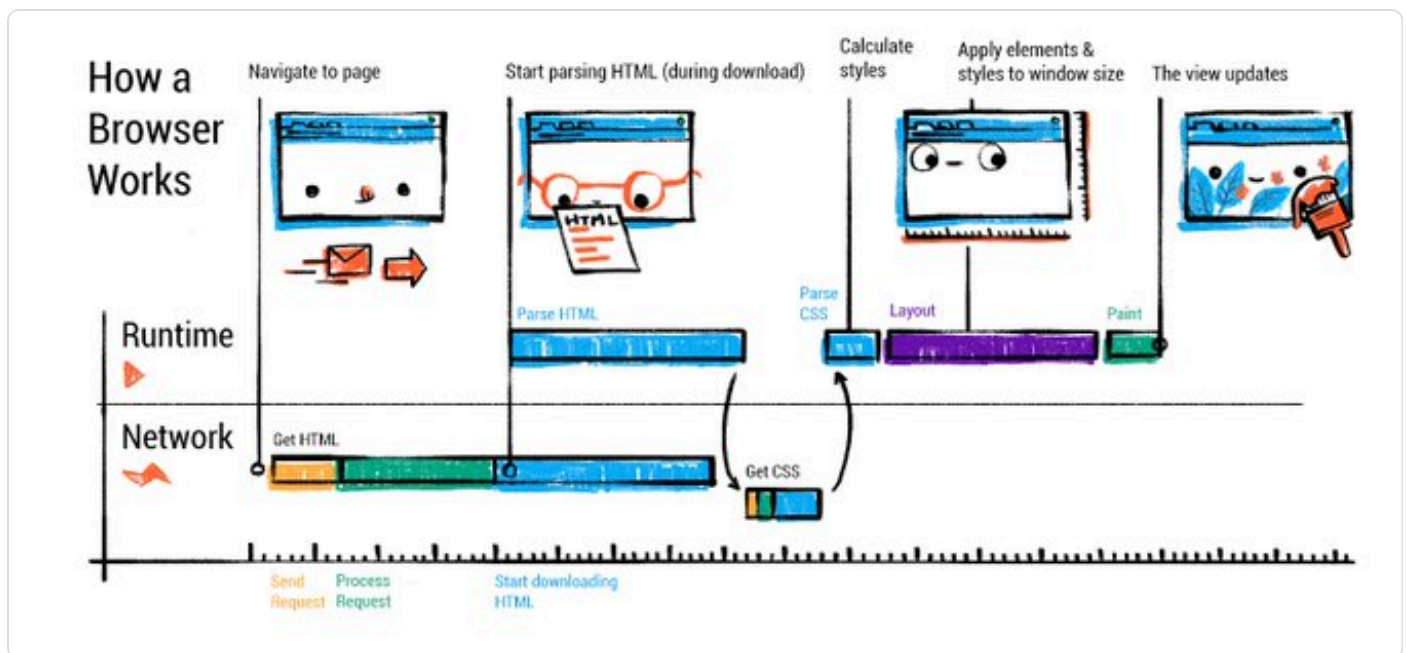


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Illustrations by Susie Lu

1. Networking and Resource Loading



Networking and resource loading pipeline — Illustration by Susie Lu

Every page load begins with the browser's networking stack fetching resources from the web. When you enter a URL or click a link, the **browser's UI thread (running in the "browser process")** kicks off a navigation request.

*The **browser process** is the main, controlling process that manages all other processes and the browser's user interface. Everything that happens outside of a specific web page tab is controlled by the browser process.*

The steps include:

Some free text>>>>

URL parsing and security checks: The browser parses the URL to determine the scheme (http, https, etc.) and target domain. It also decides if the input is a search query or URL (in Chrome's omnibox, for example). Security features like blocklists may be checked here to avoid phishing sites.

DNS lookup: The network stack resolves the domain name to an IP address (unless it's cached). This may involve contacting a DNS server. Modern browsers might use OS DNS services or even DNS over HTTPS (DoH) if configured, but ultimately they obtain an IP for the host.

Establishing a connection: If no open connection to the server exists, the browser opens one. For HTTPS URLs, this includes a TLS handshake to securely exchange keys and verify certificates. The browser's network thread handles protocols like TCP/TLS setup transparently.